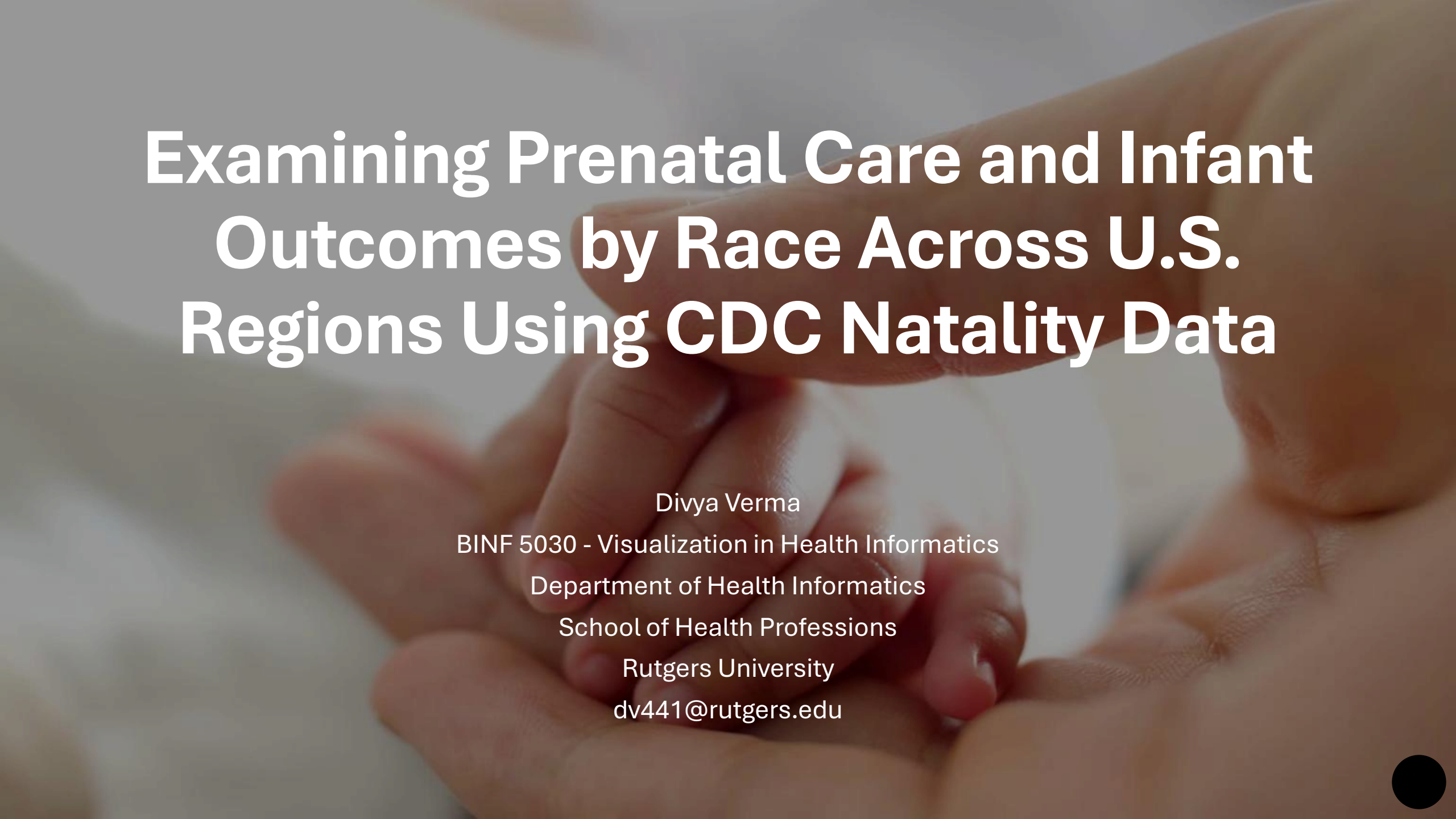




Examining Prenatal Care and Infant Outcomes by Race Across U.S. Regions Using CDC Natality Data

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Background: Why I chose this topic

Healthy People 2030 (HHS)

- Increase the proportion of pregnant women who receive early and adequate prenatal care — MICH-08: Started with 76.4% (2018) → 80.5% (2030)
- Reduce the rate of fetal deaths at 20 or more weeks of gestation — MICH-01: Started with 5.9 in 2017 → 5.7 per 1,000 live births in 2021



Infant Health Concern

- 2022: >20,000 infant deaths in the U.S. → IMR = 5.61/1,000 live births (+3% from 2021)
- 2023: Infant mortality rate unchanged; Low birthweight (LBW) prevalence = 8.58%

Key Causes & Prevention

- Pregnancy-related complications = 1 of 5 leading causes of infant mortality (CDC, 2024)
- Timely, high-quality prenatal care reduces risk of preterm birth, LBW, and postpartum depression (Thai & Johnson, 2022)

Identify and Resolve Racial and Geographic Disparities

- Highest infant mortality: Black non-Hispanic (10.93/1,000) > AI/AN (9.20) > NH/PI (8.21) > Hispanic (5.03) > White (4.48) > Asian (3.44)
- LBW prevalence: Black = 14.8%, Hispanic = 7.9%, White = 7.0%
- Neighborhood-level variation: Chicago (2010–2014) IMR 1.4 → 24.5/1,000; highest in predominantly Black communities (3.63× higher)



Research Questions

1. Variation in Prenatal Care Access & Utilization: How prenatal care and access vary by race and region across the U.S., for prenatal visits?

2. Variation in Infant Outcomes: How do infant outcomes vary by race and region across the U.S., for infant birth weight?

3. How do differences in prenatal care utilization relate to infant outcomes?

4. Which U.S. regions and races show the most significant disparities in prenatal care access and infant health outcomes?

Data Story

- Data Sources and methods
- Historical Trend 2016-2024
- 2024 Data

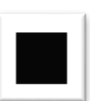
- By State → Race
- By County → Race

- By State → Race
- By County → Race

- Identify patterns and correlation by region and by races

- County level disparities among races

- Conclusion
- Actionable Insights



Data Source

- **CDC WONDER – NVSS Natality Public Use Data**
 - Maternal and infant characteristics: State, County, Race, Prenatal visits, Birth weight, Infant Survival Status
- Time span: **2016–2024**

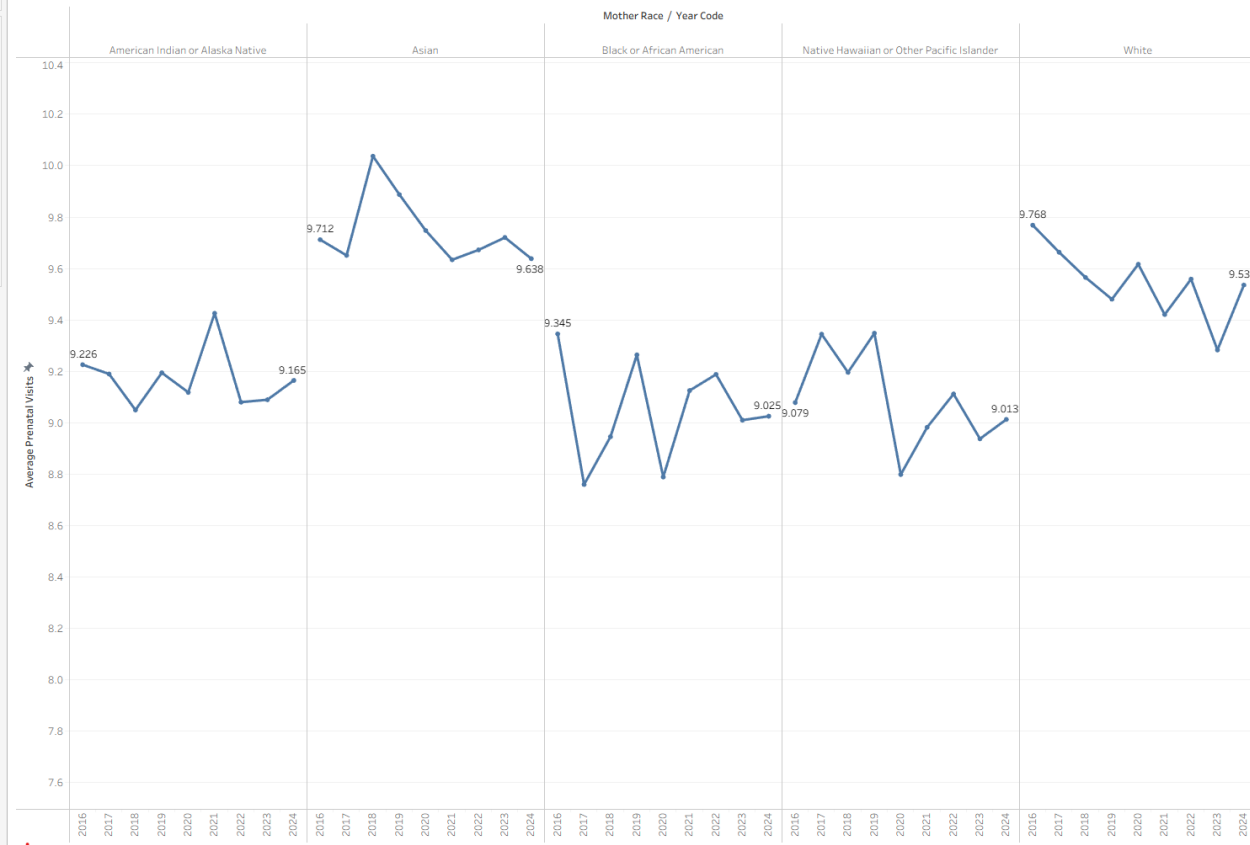
Dataset	Year	Cases	Key Measures	Purpose
1. Prenatal Visits by Race & Geography	2024	17,227	Race, #Prenatal Visits, Avg Visits, #Births, State, County	Compare prenatal care patterns by race & geography
2. Infant Birth Weight by Race & Geography	2024	13,280	Race, Birth Weight, Avg Birth Weight, Infant Mortality Rate, #Births, State, County	Examine infant outcomes by race & prenatal care
3. Prenatal Visits & Infant Birthweight by Year	2016–2024	8,503	Race, Prenatal Visits, Avg Birth Weight, Year	Examine time trends in infant birth weight by race

- **Analysis and Visualization Tool:** Tableau Prep and Tableau Desktop
- **Limitations:**
 - Infant Mortality Rate was excluded due to suppressed values(unreliable)
 - Not all variables were analyzed.
 - Limitation of rows of exported data

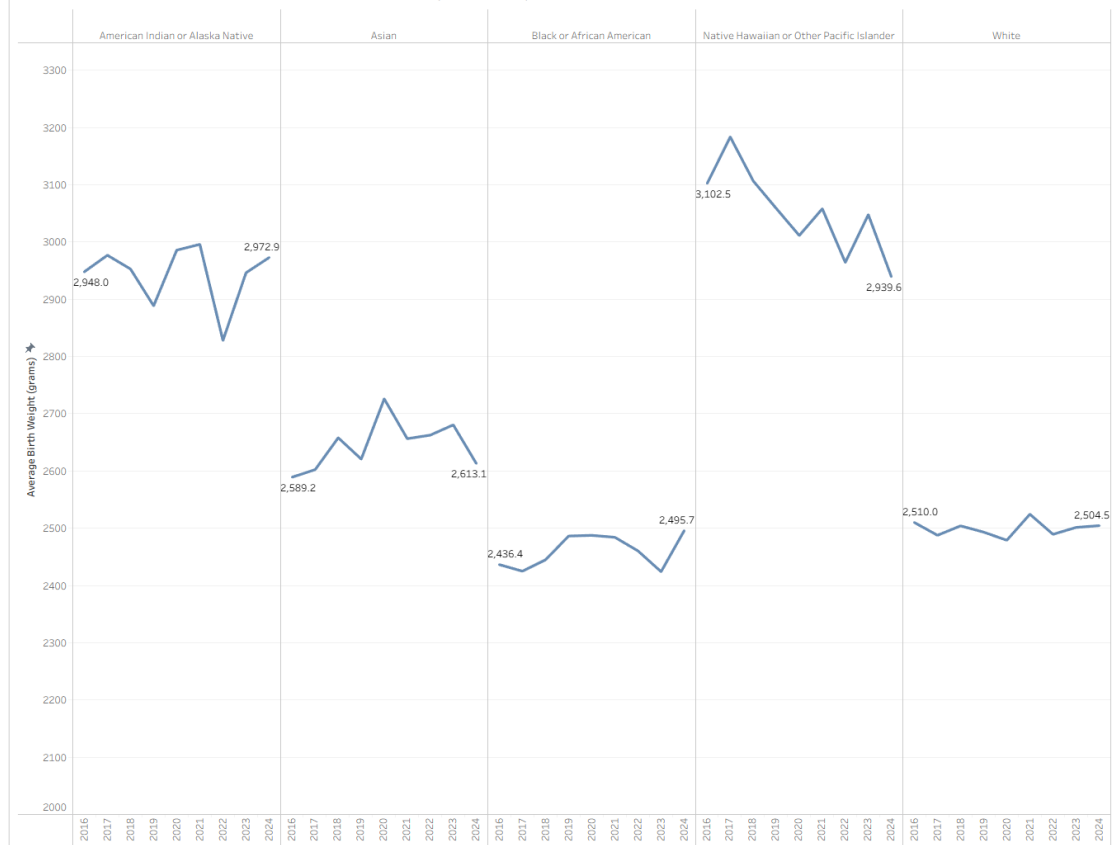


Trend over years for Prenatal Visits and Infant Birth weight 2016-2024

A. Line Chart: Average Prenatal Visits by Mother's Race (2016-2024)

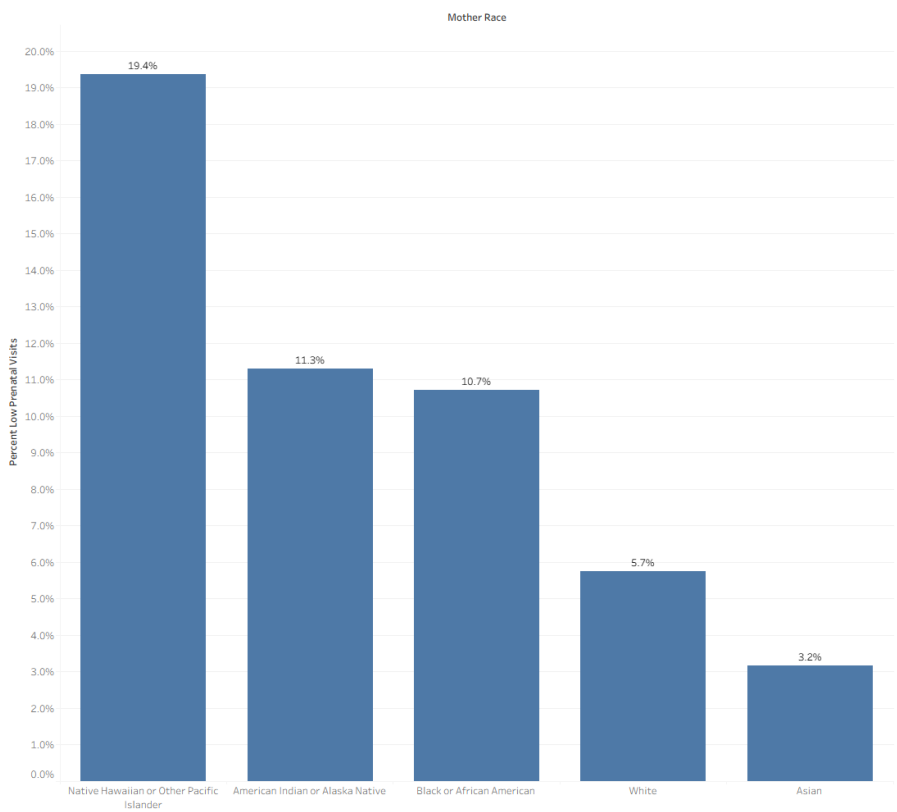


B. Line Chart: Average Birth Weight by Mother's Race (2016-2024)

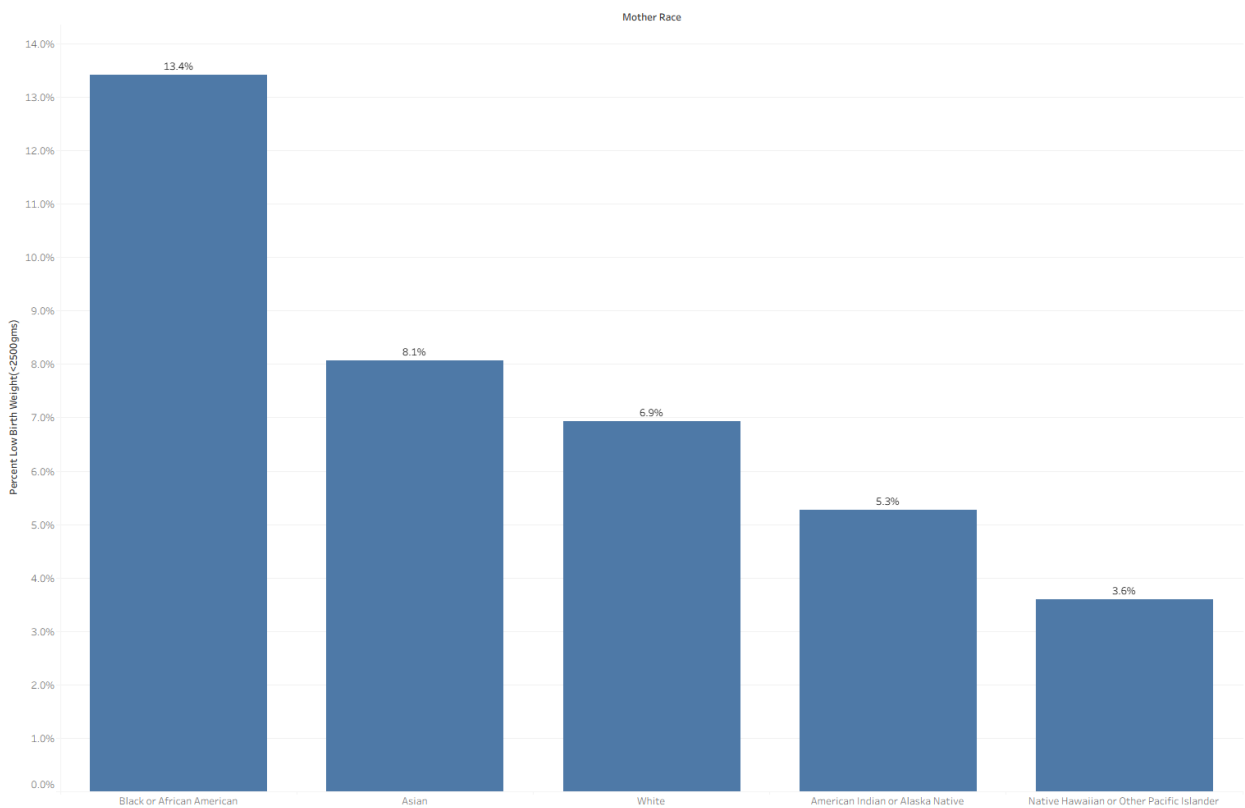


Prenatal Visits and Infant Birth weight by Race (2024)

C. Percentage of Low Prenatal Visits by Race 2024 2024



D. Percentage of Low Birth Weight by Race 2024



1. Prenatal Care Access & Utilization:

How prenatal care and access vary by race and region across the U.S., for prenatal visits?

How many prenatal visits are recommended?

- 12 to 14 in-person visits (can be fewer for average or low-risk patients)

Dataset 1

- Prenatal care visits range between No visits- 19 or more visits
- Low prenatal visits threshold set as less than <5 visits
- Weighted prenatal visits

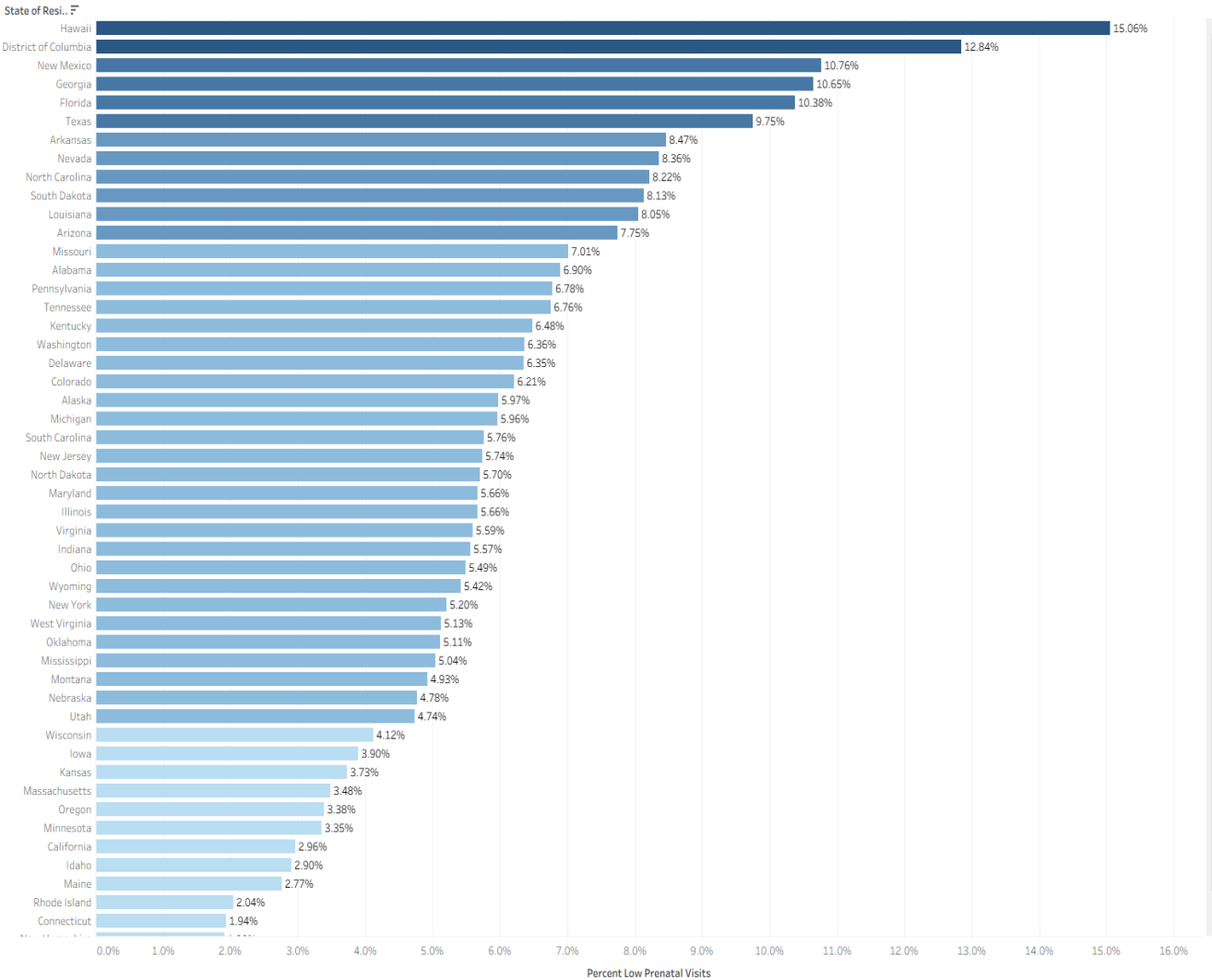
$$\% \text{Low Prenatal Visits} = \frac{\sum (1 \text{ if AvgPrenatalVisits} < 5 \times \text{Births})}{\sum (\text{Births})}$$





States with Highest Rates of Inadequate Prenatal Care (<5 Visits, weighted)-2024

1A. Bar Chart: States with Highest Rates of Inadequate Prenatal Care (<5 Visits) 2024
Hawaii, District of Columbia, New Mexico, Georgia, Florida, Texas have highest proportion of low prenatal visits.



1B. Highlight Table: States with Highest Rates of Inadequate Prenatal Care (<5 Visits) by Race 2024

- Native Hawaiian mothers** recieved lowest prenatal visits in Arkansas, Alabama, Hawaii and Missouri.
- Black mothers** recieved lowest prenatal visits in District of Columbia, South Dakota, Florida and Missouri.
- American Indian mothers** recieved lowest prenatal visits in South Dakota, North Dakota, Mississippi and Montana.

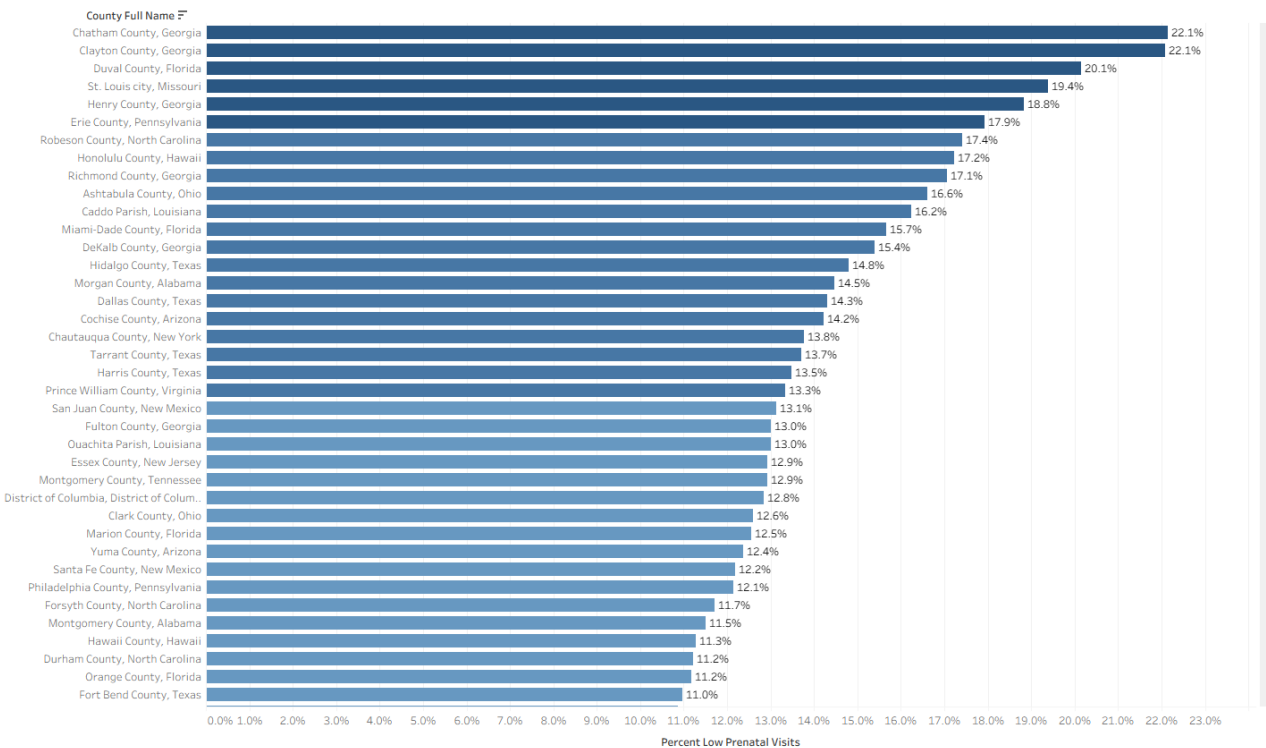
State of Resi..		Mother Race				
		American Indian or ..	Asian	Black or African Am..	Native Hawaiian or ..	White
Hawaii			16.26%	0.00%	35.91%	7.89%
District of Columbia			8.01%	17.30%		9.05%
New Mexico	16.15%	0.00%	8.49%			10.30%
Georgia	0.00%	7.70%	14.52%	11.20%		8.80%
Florida	10.17%	4.75%	15.68%			9.07%
Texas	3.51%	6.08%	13.60%	20.65%		9.44%
Arkansas	0.00%	1.77%	13.84%	49.05%		6.84%
Nevada	0.00%	4.97%	12.61%	10.67%		7.97%
North Carolina	9.88%	3.97%	10.71%	0.00%		7.62%
South Dakota	33.11%	0.00%	15.79%			3.71%
Louisiana	0.00%	0.00%	10.98%			6.48%
Arizona	13.61%	4.09%	10.20%	8.52%		7.39%
Missouri	0.00%	1.64%	15.09%	34.67%		5.82%
Alabama	16.81%	0.00%	7.71%	44.83%		6.67%
Pennsylvania	9.52%	3.66%	11.55%	0.00%		6.01%
Tennessee	4.74%	0.88%	10.81%	5.98%		6.10%
Kentucky		1.26%	9.79%	0.00%		6.21%
Washington	10.49%	2.62%	7.86%	22.60%		6.34%
Delaware		0.00%	9.23%			5.54%
Colorado	2.29%	2.38%	9.47%	25.66%		6.18%
Alaska	10.63%	2.34%	0.00%	16.03%		4.54%
Michigan	0.00%	1.65%	11.22%			5.12%
South Carolina	0.00%	0.00%	7.43%			5.27%
New Jersey	0.00%	2.36%	11.89%	0.00%		4.89%
North Dakota	28.48%	0.00%	5.47%	19.61%		3.91%
Maryland	0.00%	1.92%	7.51%	0.00%		5.18%
Illinois	0.00%	2.88%	11.52%			4.83%
Virginia	0.00%	4.31%	7.08%	0.00%		5.32%
Indiana	0.00%	3.04%	11.17%			4.77%
Ohio	0.00%	1.36%	9.62%	0.00%		4.78%
Wyoming	19.25%	0.00%	0.00%			5.03%
New York	0.00%	3.71%	9.07%	0.00%		4.42%
West Virginia		0.00%	4.87%			5.18%
Oklahoma	6.42%	0.80%	8.23%	28.57%		4.64%
Mississippi	24.78%	0.00%	7.15%			3.42%
Montana	24.13%	0.00%	0.00%			3.14%
Nebraska	8.92%	0.00%	7.02%			4.70%
Utah	3.00%	2.64%	8.69%	13.53%		4.59%
Wisconsin	8.31%	1.55%	9.80%			3.59%
Iowa	0.00%	0.00%	7.58%	26.67%		3.48%
Kansas	0.00%	0.00%	6.98%	0.00%		3.63%
Massachusetts		1.36%	7.46%			2.81%
Oregon	5.98%	0.00%	2.81%	0.00%		3.58%



Counties with Highest Rates of Low Prenatal Visits (<5 Visits, weighted)-2024

2A. Bar Chart: Counties with Highest Rates of Inadequate Prenatal Care (<5 Weighted Prenatal Visits) 2024

-Georgia had the most low-prenatal-care counties: Chatham, Clayton, Henry, Richmond, -DeKalb, and Fulton.
-Florida: Duval and Miami-Dade had the lowest prenatal visits.
-Texas: Hidalgo and Dallas counties showed the lowest levels.
-Single low-prenatal-care counties appeared in several states, including Missouri, Pennsylvania, North Carolina and Hawaii
-->showing disparities exist at county level.



2B. Highlight Table: Counties with Highest Percentage of Mothers with Low Prenatal Visits(<5) by Race 2024
-Native Hawaiian mothers had the lowest visits in Woodbury Iowa (54.55%).
-American Indian mothers had lowest prenatal visits in Duval county, Florida (52%).

County Full Name	Mother Race				
	American Indian or Alaska ..	Asian	Black or African American	Native Hawaiian or Other ..	White
Duval County, Florida	52.17%	10.99%	27.23%		16.78%
Hennepin County, Minnesota	28.21%	1.56%	9.49%		3.11%
Pennington County, South Dak..	26.77%				2.11%
Pierce County, Washington	25.73%	8.32%	11.04%	18.96%	9.55%
Minnehaha County, South Dak..	25.15%	0.00%	20.22%		4.27%
Pima County, Arizona	22.63%	3.50%	16.41%	0.00%	8.58%
San Juan County, New Mexico	22.71%				10.57%
Bernalillo County, New Mexico	14.56%	0.00%	14.88%		9.87%
Maricopa County, Arizona	13.76%	4.41%	9.90%	9.15%	6.91%
Philadelphia County, Pennsylv..	13.73%	8.26%	15.39%	0.00%	9.85%
King County, Washington	12.92%	2.33%	7.05%	19.75%	3.84%
Navajo County, Arizona	12.37%				5.18%
Spokane County, Washington	12.22%	0.00%	18.84%	51.33%	9.68%
Robeson County, North Caroli..	11.83%		21.04%		21.85%
Davidson County, Tennessee	11.76%	4.34%	12.54%	0.00%	8.65%
Yellowstone County, Montana	11.61%				3.34%
Sandoval County, New Mexico	11.02%	0.00%			7.56%
Coconino County, Arizona	9.05%				4.49%
Tulsa County, Oklahoma	8.09%	0.00%	10.72%	22.22%	5.43%
Harris County, Texas	7.91%	6.33%	14.75%	18.18%	13.80%
Oklahoma County, Oklahoma	7.39%	0.00%	8.72%		4.38%
Salt Lake County, Utah	7.33%	4.23%	13.26%	17.66%	6.06%
Multnomah County, Oregon	5.85%	0.00%	3.70%	0.00%	3.21%
Yakima County, Washington	5.81%	0.00%	0.00%		5.04%
Anchorage Borough, Alaska	5.35%	3.28%	0.00%	16.80%	4.22%
Riverside County, California	5.13%	4.17%	4.33%	0.00%	3.48%
Pinal County, Arizona	4.76%	0.00%	4.50%		4.99%
Los Angeles County, California	3.43%	3.35%	5.81%	0.00%	2.76%
Yuma County, Arizona	0.00%	0.00%	0.00%		12.60%
Yavapai County, Arizona	0.00%	0.00%			4.60%
Woodbury County, Iowa	0.00%		0.00%	54.55%	4.81%
Williamson County, Texas	0.00%	9.02%	9.89%		6.27%
Whatcom County, Washington	0.00%	0.00%	0.00%		6.02%
Westchester County, New York	0.00%	1.72%	5.61%		2.33%
Weld County, Colorado	0.00%	8.97%	17.07%		8.28%
Weber County, Utah	0.00%	0.00%	0.00%		6.00%
Wayne County, Michigan	0.00%	1.33%	12.87%		5.34%
Washoe County, Nevada	0.00%	5.68%	14.21%	30.30%	8.14%
Washington County, Oregon	0.00%	0.00%	4.21%		2.95%
Washington County, Arkansas	0.00%	0.00%	10.54%	49.26%	7.38%
Wake County, North Carolina	0.00%	4.79%	15.51%		8.58%
Utah County, Utah	0.00%	0.00%	0.00%	7.41%	3.80%

County Full Name	Mother Race				
	American Indian or Alaska ..	Asian	Black or African American	Native Hawaiian or Other ..	White
Greene County, Missouri		0.00%	29.58%		9.59%
Duval County, Florida	52.17%	10.99%	27.23%		16.78%
Berrien County, Michigan		0.00%	26.97%		5.56%
St. Louis city, Missouri		0.00%	25.35%		15.01%
Clark County, Ohio			23.62%		8.12%
Chatham County, Georgia		17.65%	22.99%		21.73%
Erie County, Pennsylvania		0.00%	22.94%		17.86%

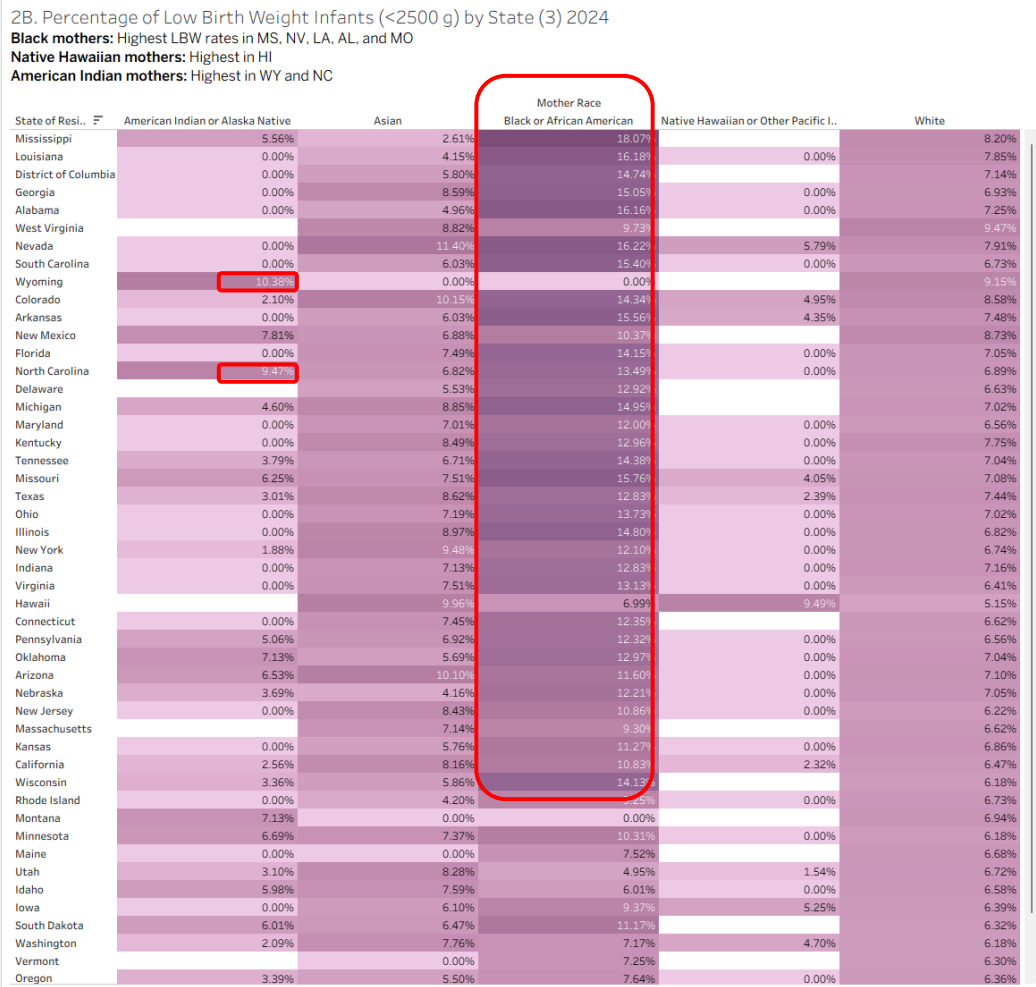
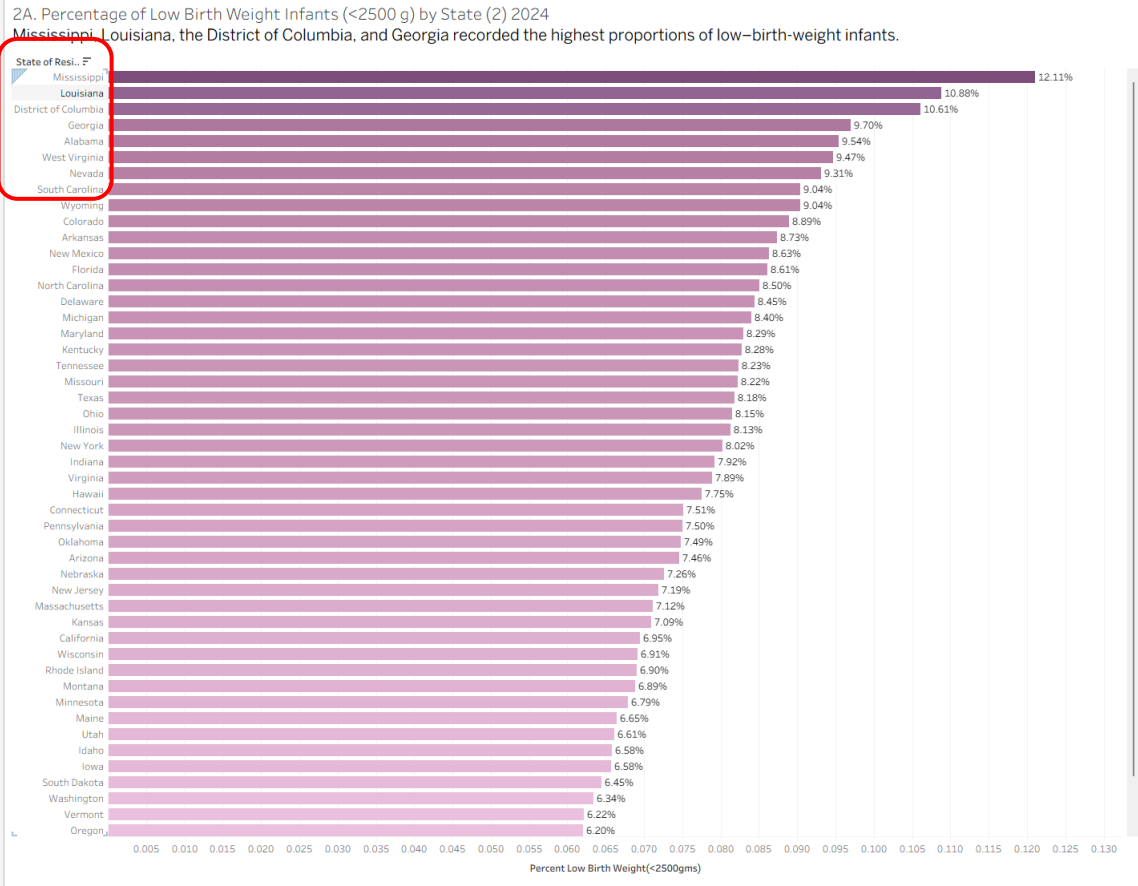
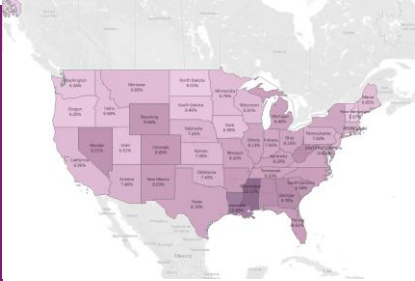
2. Variation in Infant Outcomes: How do infant outcomes vary by race and region across the U.S., for infant birth weight?

- Dataset 2- Average Low Birth Weight Range= 2800 grams -3200 grams
- County Weighted Birth Weight
- Low Birth Weight Infants <2500 grams were flagged → Weighted Low Birth Weight Count → Percent Low Birth Weight

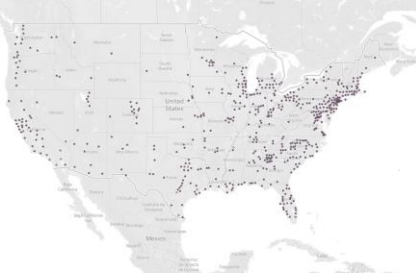
$$\text{County Weighted Birth Weight} = \frac{\sum(\text{AverageBirthWeight} \times \text{Births})}{\sum(\text{Births})}$$



Percentage of Low-Birth-Weight Infants (<2500 g) by State

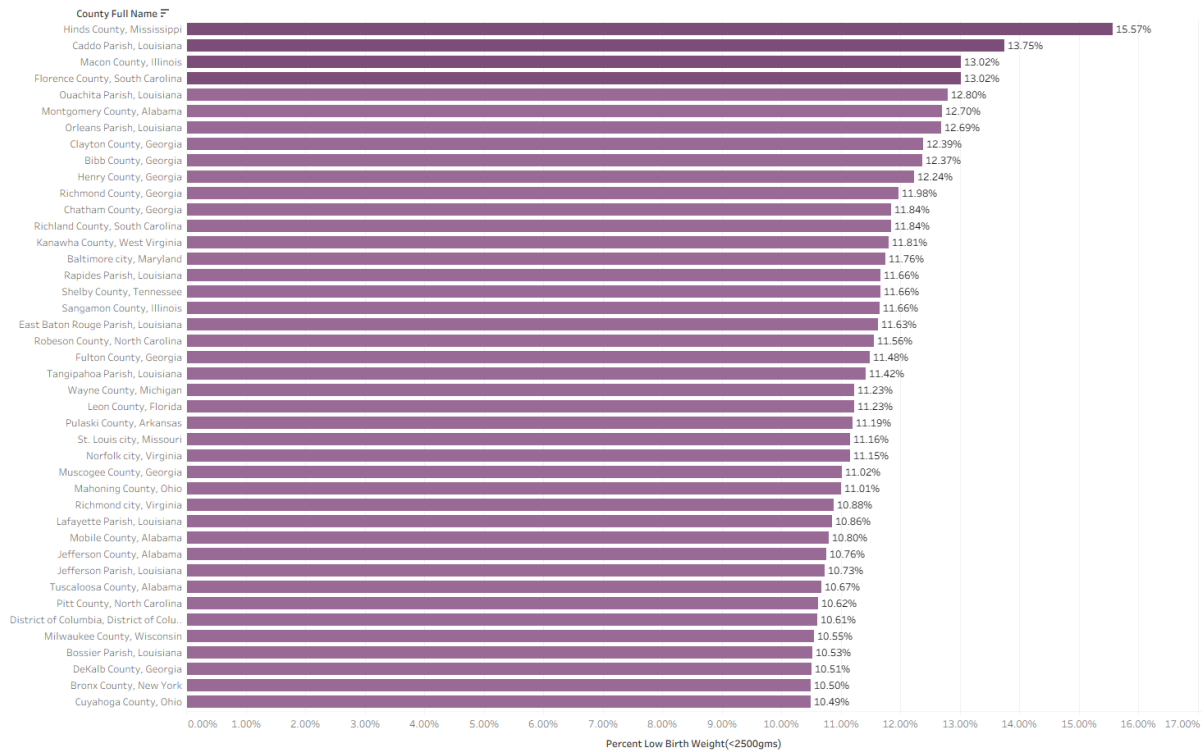


Percentage of Low-Birth-Weight Infants (<2500 g) by County



2D. Counties with Highest Percentage of Mothers with Low Birth Weights(<2500gms) in 2024

-Mississippi had the highest percentages of LBW infants in **Hinds County, MS (15.57%)**.
-Multiple **Louisiana** counties had most LBW infants- **Caddo (13.75%)**, **Ouachita (12.80%)**, **Orleans (12.69%)**, and **Rapides (11.76%)**.
-**Georgia** had several counties with high percentages of LBW infants, including **Clayton (12.39%)**, **Bibb (12.37%)**, **Henry (12.24%)**, **Richmond (11.98%)**, and **Chatham (11.84%)**.
-Other notable counties with elevated low birth weight rates included **Macon County, IL (13.02%)**, **Florence County, SC (13.02%)**, **Richland County, SC (11.84%)**, **Kanawha County, WV (11.81%)**, **Baltimore City, MD (11.76%)**, and **Shelby County, TN (11.66%)**.



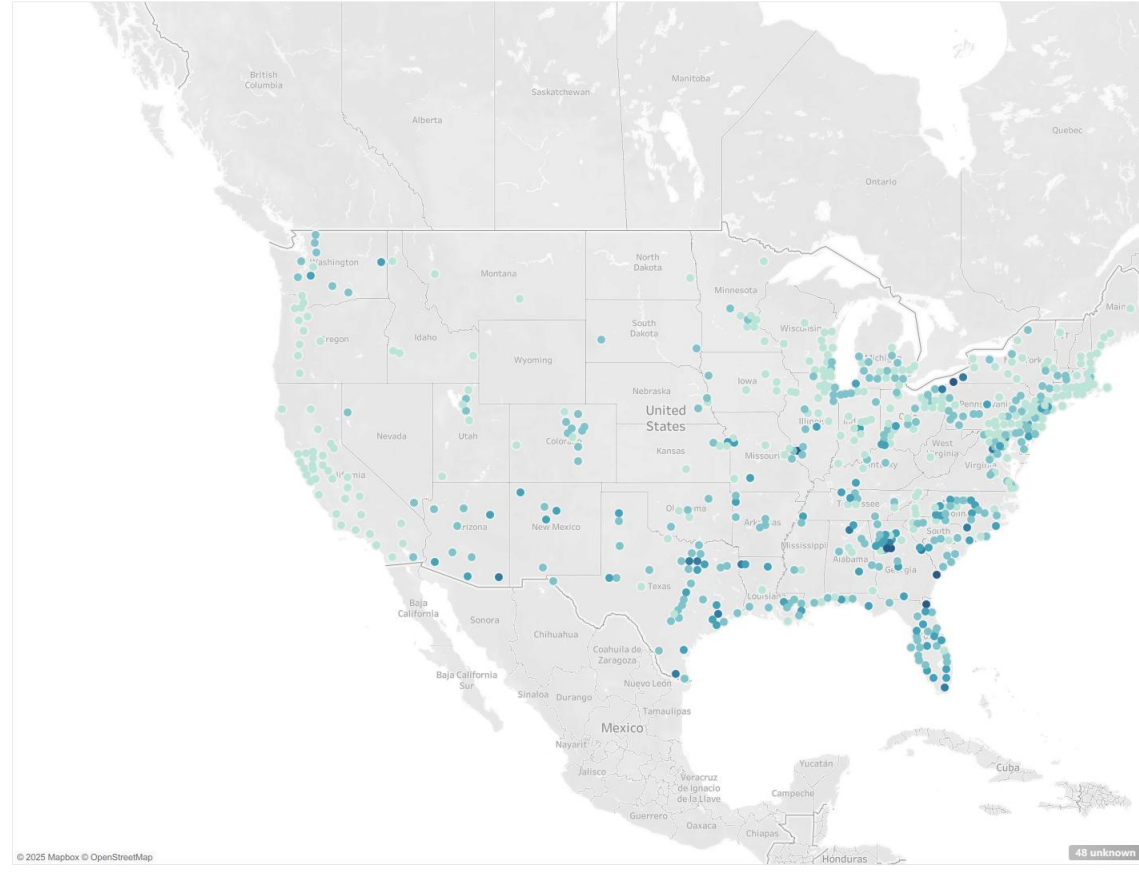
2E. Counties with Highest Percentage of Mothers with Low Birth Weights(<2500gms) by Race 2024 2024

-**Black or African American** mothers had the highest percentages of low birth weight infants, notably in **Hinds County, MS (19.27%)**, **Caddo Parish, LA (18.04%)**, and **Macon County, IL (20.88%)**.
-**Native Hawaiian** mothers showed elevated low birth weight percentages in **Harris County, Texas (13.48%)**.
-**American Indian** mothers had higher rates in Robison County, **North Carolina** counties (13.45%).
-**Asian** mothers had the lowest birth weights in Harris County, Mississippi (19%).
-**White** mothers generally had lower percentages, with the highest around 10–12% in certain counties such as West Virginia and Illinois.

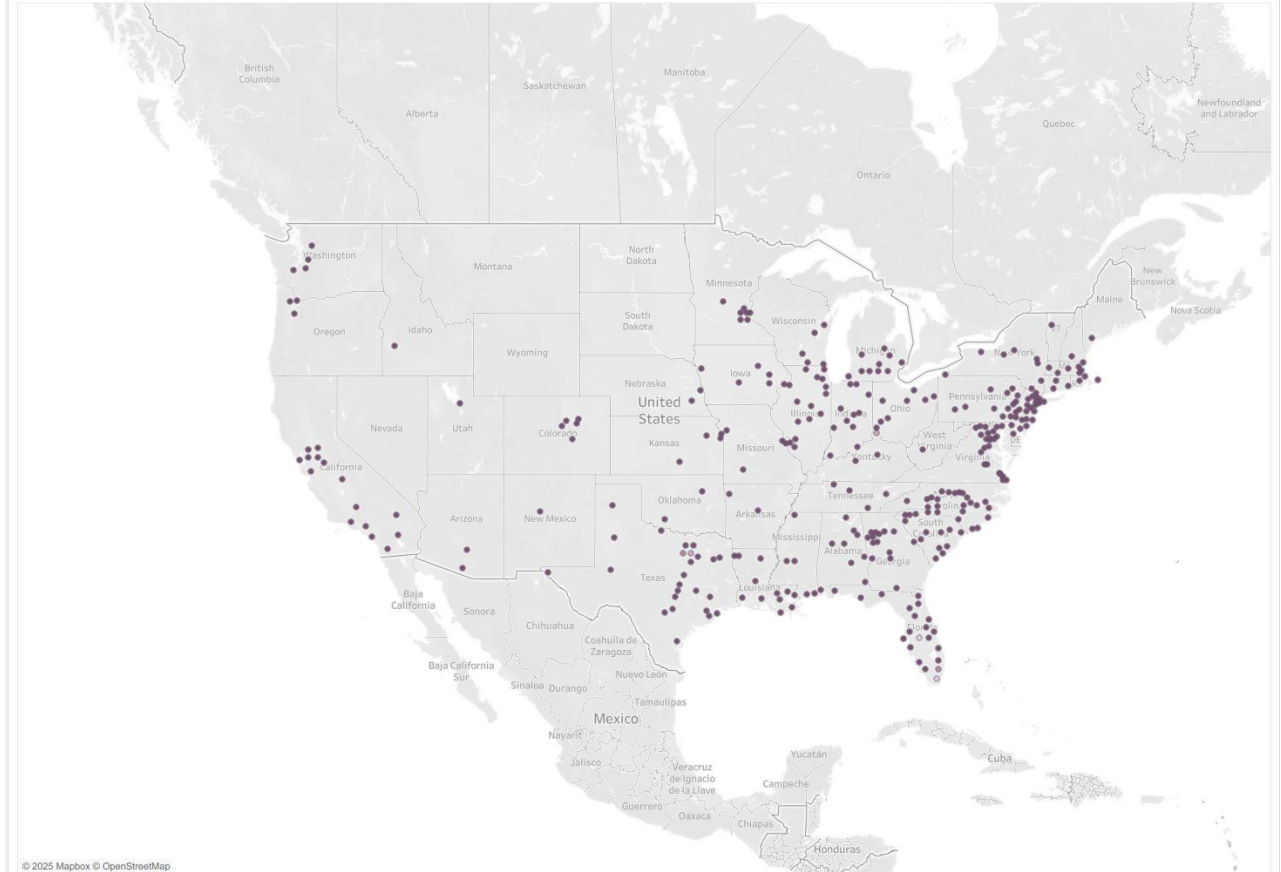
County Full Name	American Indian..	Mother Race			
		Asian	Black or African ..	Native Hawaiian..	White
Hinds County, Mississippi		0.00%	19.27%		5.52%
Caddo Parish, Louisiana		0.00%	18.04%		7.87%
Macon County, Illinois		0.00%	20.88%		10.34%
Florence County, South Carolina		0.00%	18.15%		8.93%
Ouachita Parish, Louisiana		0.00%	16.67%		9.16%
Montgomery County, Alabama		0.00%	16.62%		7.56%
Orleans Parish, Louisiana		0.00%	17.24%		6.48%
Clayton County, Georgia		7.81%	15.23%		5.81%
Bibb County, Georgia		0.00%	15.99%		6.90%
Henry County, Georgia		0.00%	14.75%		8.19%
Richmond County, Georgia		0.00%	15.94%		7.07%
Chatham County, Georgia		0.00%	18.76%		7.31%
Richland County, South Carolina	0.00%	7.81%	16.13%	0.00%	6.67%
Kanawha County, West Virginia			8.62%		12.06%
Baltimore city, Maryland	0.00%	7.30%	15.69%	0.00%	7.28%
Rapides Parish, Louisiana		0.00%	16.06%		8.91%
Shelby County, Tennessee	0.00%	7.37%	15.83%	0.00%	7.11%
Sangamon County, Illinois		0.00%	21.66%		8.55%
East Baton Rouge Parish, Louisiana		6.63%	15.97%	0.00%	6.77%
Robeson County, North Carolina	13.45%		14.18%		7.22%
Fulton County, Georgia		9.29%	16.17%		6.66%
Tangipahoa Parish, Louisiana			15.86%		8.68%
Wayne County, Michigan	0.00%	10.43%	16.66%		7.58%
Leon County, Florida		9.09%	15.93%		7.84%
Pulaski County, Arkansas	0.00%	6.41%	17.44%		6.74%
St. Louis city, Missouri		0.00%	16.41%		7.23%
Norfolk city, Virginia		0.00%	16.59%		7.98%
Muscogee County, Georgia		14.49%	14.14%		7.54%
Mahoning County, Ohio			17.09%		9.16%
Richmond city, Virginia		0.00%	17.01%		6.86%
Lafayette Parish, Louisiana	0.00%	0.00%	16.25%		8.56%
Mobile County, Alabama	0.00%	0.00%	16.30%		7.56%
Jefferson County, Alabama		6.99%	16.37%		7.09%
Jefferson Parish, Louisiana	0.00%	5.20%	16.36%		8.12%
Tuscaloosa County, Alabama		0.00%	17.56%		6.66%
Pitt County, North Carolina		0.00%	15.85%		6.37%
District of Columbia, District of Columbia	0.00%	5.80%	14.74%		7.14%
Milwaukee County, Wisconsin	0.00%	9.07%	16.77%		7.03%
Bossier Parish, Louisiana		0.00%	15.75%		8.55%
DeKalb County, Georgia		11.62%	13.99%		6.97%
Bronx County, New York	9.71%	11.32%	11.90%	0.00%	8.76%
Cuyahoga County, Ohio		7.43%	15.67%		7.76%
Lowndes County, Georgia		0.00%	16.88%		5.85%
Ascension Parish, Louisiana		0.00%	14.88%		8.69%

3. How do differences in prenatal care utilization relate to infant outcomes?

1C. Choropleth Map: Counties with Highest Rates of Inadequate Prenatal Care (<5 Weighted Prenatal Visits) 2024



2C. Percentage of Low Birth Weight Infants (<2500 g) by County 2024



- 1) Counties with poor prenatal care utilization generally correspond to higher percentages of low-birth-weight infants.
- 2) Southern U.S. bears the highest burden, highlighting regional health inequities.
- 3) While correlation is evident, causation cannot be confirmed without further statistical analysis (e.g., controlling for maternal age, race, income, and other risk factors).

4. Which U.S. regions and races show the most significant disparities in prenatal care access and infant health outcomes?

County-wise comparison of Low Prenatal Visits and Low Infant Birth Weight

1E. Highlight Table: Counties with Highest Percentage of Mothers with Low Prenatal Visits(<5) by Race 2024

-Native Hawaiian mothers had maximum proportion of <5 prenatal visits in Woodbury Iowa (54.55%), Spokane County, Washington (51.35%), Washington County, Arkansas (49.26%), Berrien County, Michigan (48.9%)
-American Indian mothers had lowest prenatal visits in Duval county, Florida (52%)

County Full Name	Mother Race				
	American Indian or Alaska Native	Asian	Black or African American	Native Hawaiian or Other Pacific Islander	White
Berrien County, Michigan		0.00%	26.97%		5.56%
Caddo Parish, Louisiana		0.00%	20.99%		9.50%
Chatham County, Georgia		17.65%			21.73%
Clark County, Ohio			23.62%		8.12%
Clark County, Washington	0.00%	0.00%	0.00%	35.44%	2.71%
Clayton County, Georgia		25.25%	20.75%		25.16%
Denver County, Colorado	0.00%	6.02%	10.13%	34.52%	7.62%
Duval County, Florida	52.17%	10.99%	27.23%		16.78%
Erie County, Pennsylvania		0.00%	22.94%		17.86%
Greene County, Missouri		0.00%	29.58%		9.59%
Hawaii County, Hawaii		0.00%		29.24%	9.31%
Honolulu County, Hawaii		19.12%	0.00%	37.87%	8.69%
Kenton County, Kentucky		0.00%	20.67%		9.99%
Miami-Dade County, Florida		19.05%	20.27%		14.67%
Minnehaha County, South Dakota	25.15%	0.00%	20.22%		4.27%
Pennington County, South Dakota	26.77%				2.11%
Richmond County, Georgia		0.00%	20.12%		13.09%
Robeson County, North Carolina	11.83%		21.04%		21.85%
St. Louis city, Missouri		0.00%	25.35%		15.01%
Washington County, Arkansas	0.00%	0.00%	10.57%	49.26%	7.38%
Woodbury County, Iowa	0.00%		0.00%	54.55%	4.81%

2E. Counties with Highest Percentage of Mothers with Low Birth Weights(<2500gms) by Race 2024 (2) 2024

-Black or African American mothers had the highest percentages of low birth weight infants, notably in Hinds County, MS (18.27%), Caddo Parish, LA (18.04%), and Mason County, IL (18.04%)
-Native Hawaiian mothers showed elevated low birth weight percentages in Harris County, Texas (13.48%)
-American Indian mothers had higher rates in Robeson County, North Carolina counties (13.45%)
-Asian mothers had the lowest birth weights in Harris County, Mississippi (19%)
-White mothers generally had lower percentages, with the highest around 10-12% in certain counties such as West Virginia and Illinois.

County Full Name	Mother Race				
	American Indian or Alaska Native	Asian	Black or African American	Native Hawaiian or Other Pacific Islander	White
Berrien County, Michigan			11.74%		6.24%
Caddo Parish, Louisiana		0.00%	18.04%		7.87%
Chatham County, Georgia		0.00%	18.76%		7.31%
Clark County, Ohio			7.66%		6.14%
Clark County, Washington	0.00%	5.72%	0.00%	7.98%	5.90%
Clayton County, Georgia		7.81%	15.23%		5.81%
Denver County, Colorado	0.00%	9.49%	15.32%	9.90%	8.63%
Duval County, Florida	0.00%	9.43%	15.69%	0.00%	6.59%
Erie County, Pennsylvania		0.00%	15.72%		7.64%
Greene County, Missouri		0.00%	12.67%		7.69%
Hawaii County, Hawaii		5.05%		9.00%	4.75%
Honolulu County, Hawaii		10.39%	6.99%	10.75%	6.91%
Kenton County, Kentucky		0.00%	0.00%		5.67%
Miami-Dade County, Florida	0.00%	6.05%	13.97%	0.00%	6.42%
Minnehaha County, South Dakota	0.00%	0.00%	9.82%		6.49%
Pennington County, South Dakota	7.63%				5.89%
Richmond County, Georgia		0.00%	15.94%		7.07%
Robeson County, North Carolina	13.45%		14.18%		7.22%
St. Louis city, Missouri		0.00%	16.41%		7.23%
Washington County, Arkansas	0.00%	0.00%	7.94%	7.38%	6.50%
Woodbury County, Iowa	0.00%	0.00%	12.17%	0.00%	7.28%

Mother Race

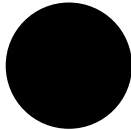
American Indian or Alaska Native

Asian

Black or African American

Native Hawaiian or Other Pacific Isla..

White



Conclusions: Prenatal Care and Infant Outcomes

- **Geographic and racial variation in prenatal care:** Low prenatal visits concentrated in the South; additional counties in West and Midwest; disparities by race (e.g., Black, Native Hawaiian, American Indian mothers).
- **Infant outcomes:** Highest low birth weight (LBW) rates in Southern counties; Black mothers show highest LBW, especially in Southeast.
- **Patterns and correlation:** Counties with poor prenatal care generally have higher LBW, suggesting a potential link.
- **Implications:** Southern U.S. shows regional and racial health inequities; correlation evident, but causation not confirmed.



Thank you

Questions?
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